Laporan Praktikum Hak Akses MySQL

Pertemuan 2: Pengelolaan Hak Akses Dosen dan Mahasiswa

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1. Tujuan

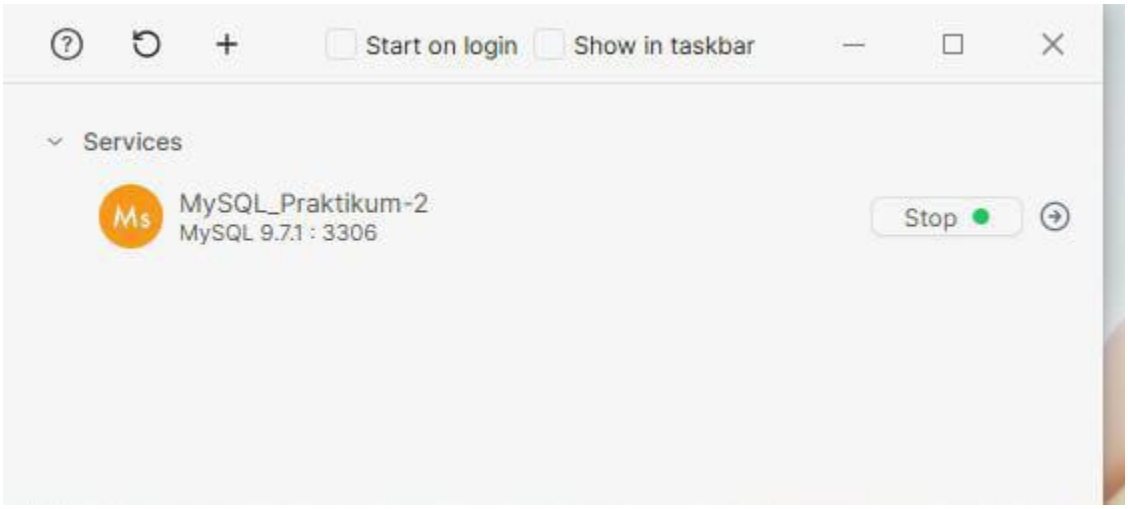
Membuat user MySQL untuk dosen dan mahasiswa, memberikan hak akses yang berbeda, menguji perintah yang boleh dan tidak boleh dijalankan, serta mencabut hak akses menggunakan REVOKE.

2. Alat

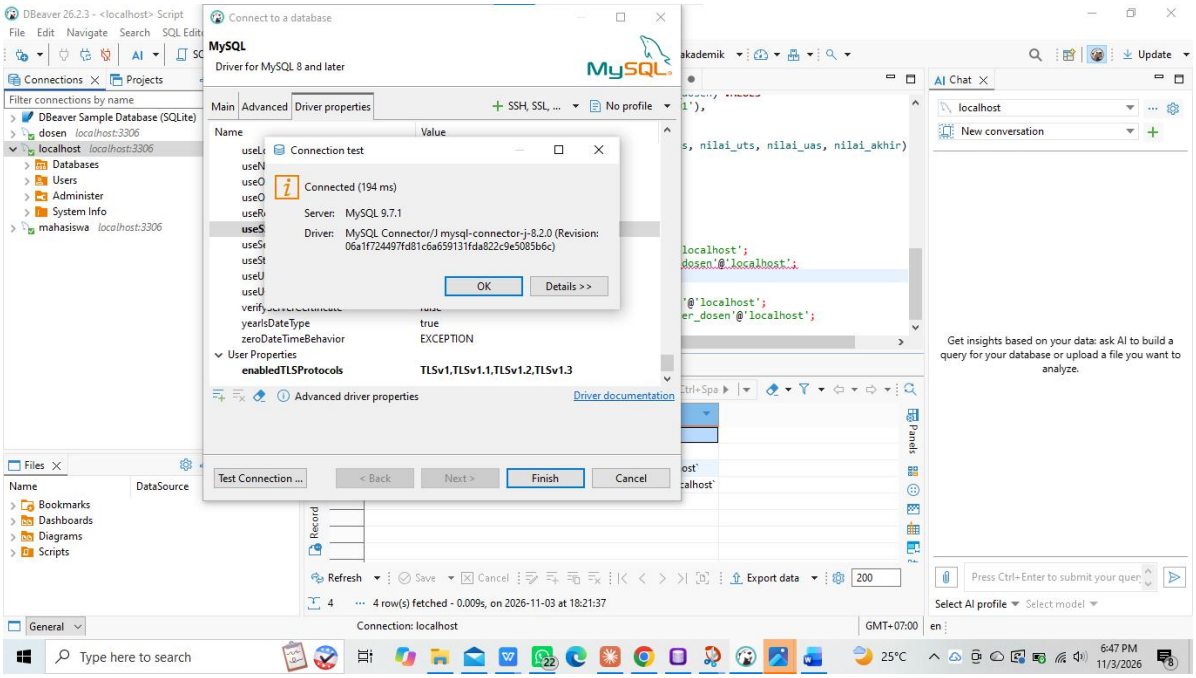
MySQL (DBngin, port 3306), DBeaver, database `akademik`.

3. Setup MySQL dan Koneksi DBeaver

DBngin: menambahkan service MySQL bernama `MySQL_Praktikum-2` (port 3306), lalu klik Start hingga status Running.



DBeaver: membuat koneksi baru ke MySQL dengan host `localhost`, port `3306`, username `root`, lalu Test Connection hingga berhasil.



4. Membuat Database dan Tabel

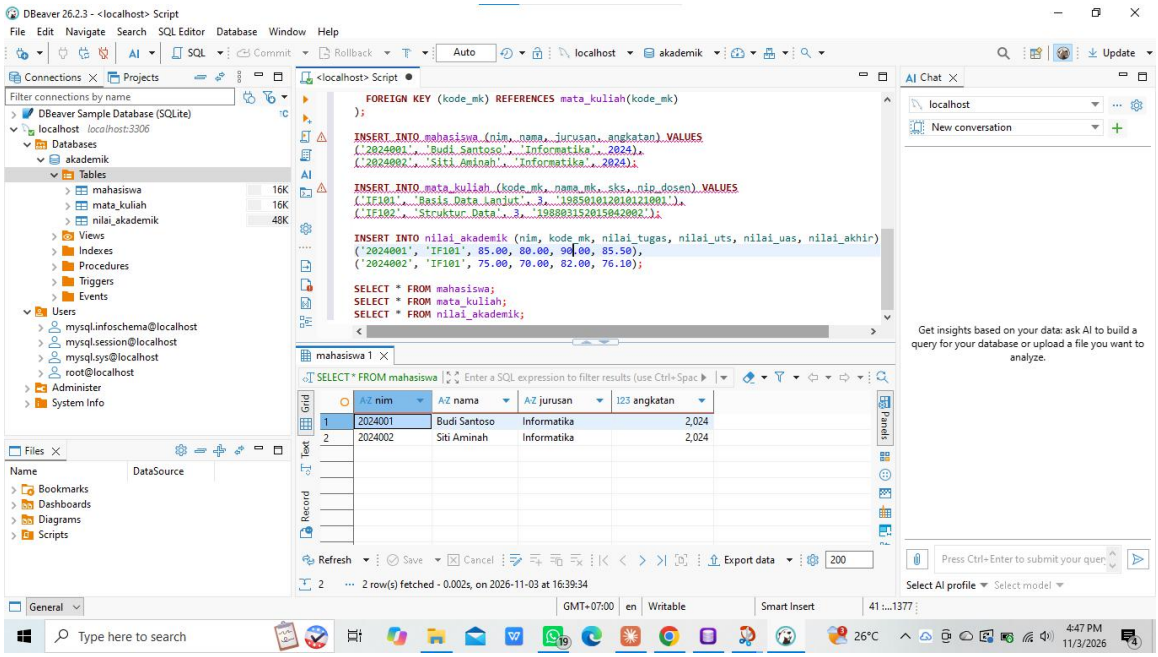
```
CREATE DATABASE IF NOT EXISTS akademik;
USE akademik;

CREATE TABLE IF NOT EXISTS mahasiswa (
    nim VARCHAR(10) PRIMARY KEY,
    nama VARCHAR(100) NOT NULL,
    jurusan VARCHAR(50) NOT NULL,
    angkatan INT NOT NULL
);

CREATE TABLE IF NOT EXISTS mata_kuliah (
    kode_mk VARCHAR(10) PRIMARY KEY,
    nama_mk VARCHAR(100) NOT NULL,
    sks INT NOT NULL CHECK (sks > 0),
    nip_dosen VARCHAR(18) NOT NULL
);

CREATE TABLE IF NOT EXISTS nilai_akademik (
    id_nilai INT AUTO_INCREMENT PRIMARY KEY,
    nim VARCHAR(10),
    kode_mk VARCHAR(10),
    nilai_tugas DECIMAL(5,2) CHECK (nilai_tugas BETWEEN 0 AND 100),
    nilai_uts DECIMAL(5,2) CHECK (nilai_uts BETWEEN 0 AND 100),
    nilai_uas DECIMAL(5,2) CHECK (nilai_uas BETWEEN 0 AND 100),
    nilai_akhir DECIMAL(5,2),
    FOREIGN KEY (nim) REFERENCES mahasiswa(nim),
    FOREIGN KEY (kode_mk) REFERENCES mata_kuliah(kode_mk)
);
```

Hasil: database akademik dan tiga tabel (mahasiswa, mata_kuliah, nilai_akademik) berhasil dibuat.



5. Mengisi Data Awal

```
INSERT INTO mahasiswa (nim, nama, jurusan, angkatan) VALUES
('2024001', 'Budi Santoso', 'Informatika', 2024),
('2024002', 'Siti Aminah', 'Informatika', 2024);

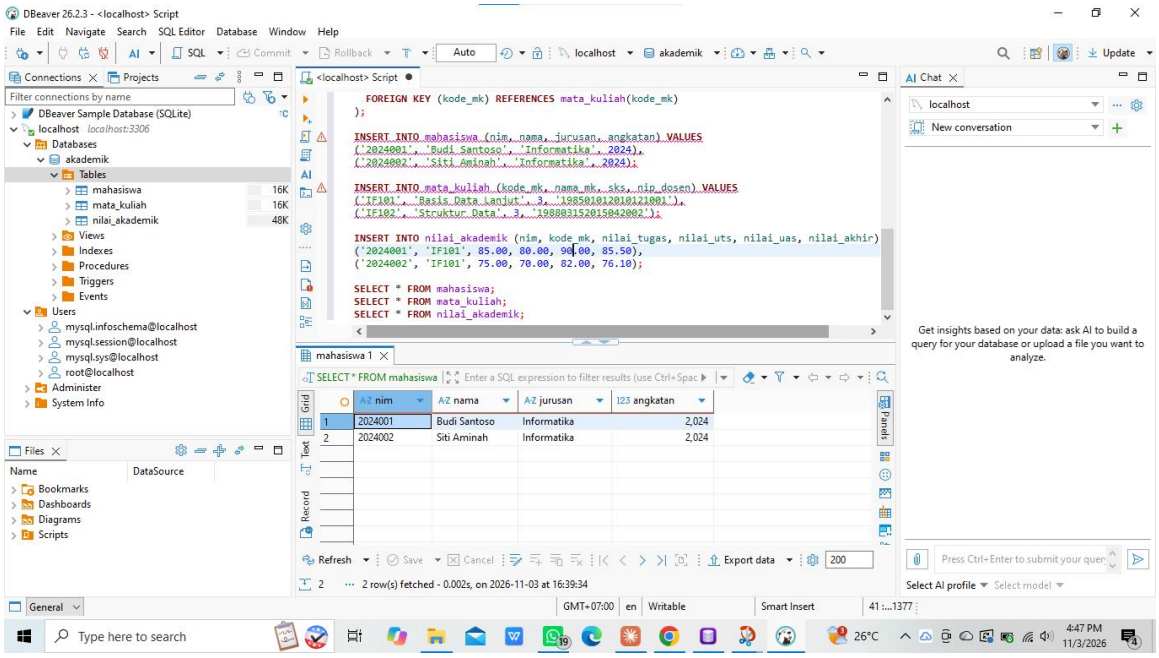
INSERT INTO mata_kuliah (kode_mk, nama_mk, sks, nip_dosen) VALUES
('IF101', 'Basis Data Lanjut', 3, '198501012010121001'),
('IF102', 'Struktur Data', 3, '198803152015042002');

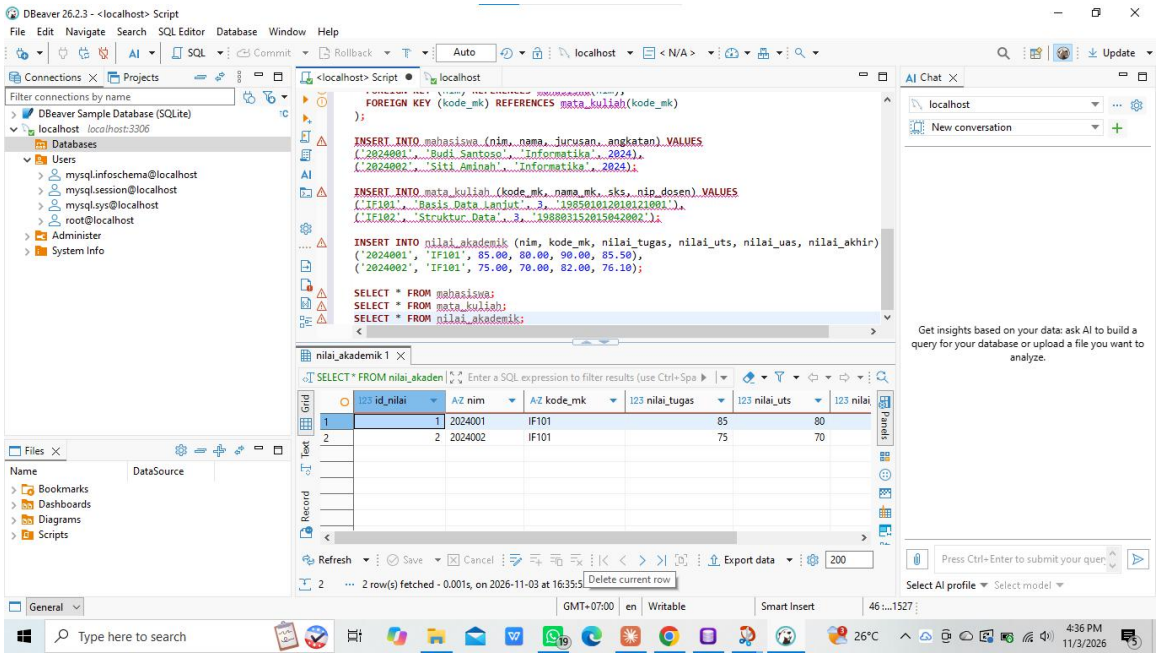
INSERT INTO nilai_akademik (nim, kode_mk, nilai_tugas, nilai_uts, nilai_uas,
nilai_akhir) VALUES
('2024001', 'IF101', 85.00, 80.00, 90.00, 85.50),
('2024002', 'IF101', 75.00, 70.00, 82.00, 76.10);
```

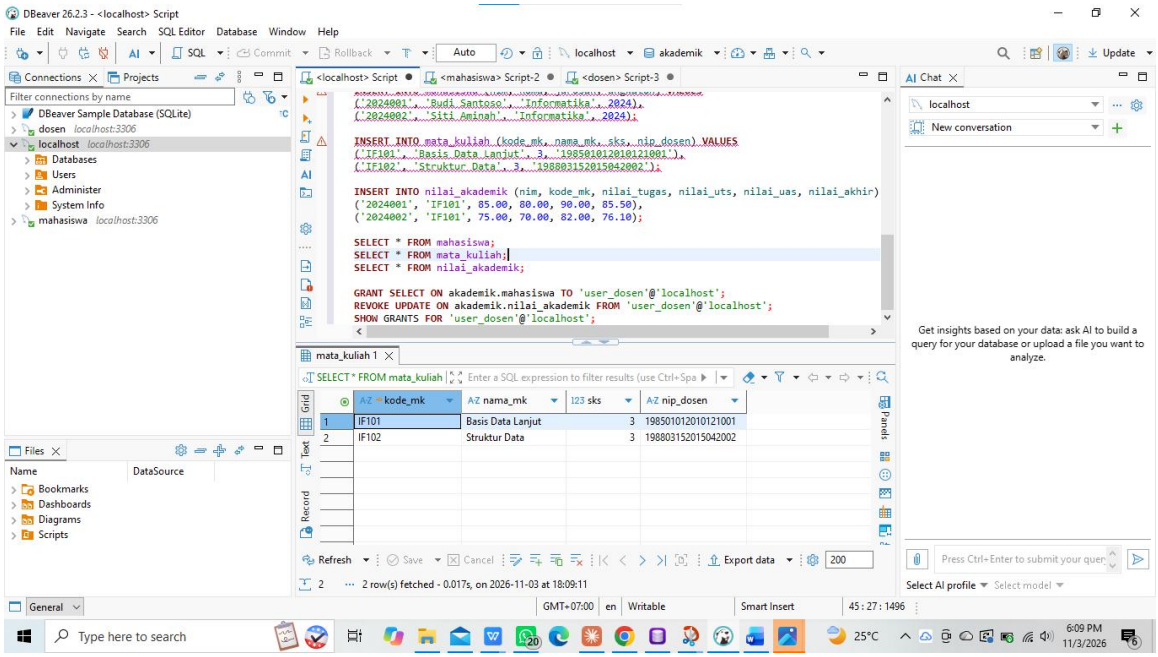
6. Mengecek Data

```
SELECT * FROM mahasiswa;
SELECT * FROM mata_kuliah;
SELECT * FROM nilai_akademik;
```

Hasil: masing-masing tabel menampilkan data yang telah dimasukkan (2 baris).


DBeaver 26.2.3 - <localhost> Script
File Edit Navigate Search SQL Editor Database Window Help
FOREIGN KEY (kode_mk) REFERENCES mata_kuliah(kode_mk)
);
INSERT INTO mahasiswa (nim, nama, jurusan, angkatan) VALUES
('2024001', 'Budi Santoso', 'Informatika', 2024),
('2024002', 'Siti Aminah', 'Informatika', 2024);
INSERT INTO mata_kuliah (kode_mk, nama_mk, sks, nip_dosen) VALUES
('IF101', 'Basis Data Lanjut', 3, '198501012010121001'),
('IF102', 'Struktur Data', 3, '198803152015042002');
INSERT INTO nilai_akademik (nim, kode_mk, nilai_tugas, nilai_uts, nilai_uas, nilai_akhir)
('2024001', 'IF101', 85.00, 80.00, 90.00, 85.50),
('2024002', 'IF101', 75.00, 70.00, 82.00, 76.10);
SELECT * FROM mahasiswa;
SELECT * FROM mata_kuliah;
SELECT * FROM nilai_akademik;
mahasiswa 1
nim nama jurusan angkatan
2024001 Budi Santoso Informatika 2,024
2024002 Siti Aminah Informatika 2,024
2 row(s) fetched - 0.002s, on 2026-11-03 at 16:39:34
AI Chat
Get insights based on your data: ask AI to build a query for your database or upload a file you want to analyze.
4:47 PM
11/3/2026


DBeaver 26.2.3 - <localhost> Script
File Edit Navigate Search SQL Editor Database Window Help
FOREIGN KEY (kode_mk) REFERENCES mata_kuliah(kode_mk)
);
INSERT INTO mahasiswa (nim, nama, jurusan, angkatan) VALUES
('2024001', 'Budi Santoso', 'Informatika', 2024),
('2024002', 'Siti Aminah', 'Informatika', 2024);
INSERT INTO mata_kuliah (kode_mk, nama_mk, sks, nip_dosen) VALUES
('IF101', 'Basis Data Lanjut', 3, '198501012010121001'),
('IF102', 'Struktur Data', 3, '198803152015042002');
INSERT INTO nilai_akademik (nim, kode_mk, nilai_tugas, nilai_uts, nilai_uas, nilai_akhir)
('2024001', 'IF101', 85.00, 80.00, 90.00, 85.50),
('2024002', 'IF101', 75.00, 70.00, 82.00, 76.10);
SELECT * FROM mahasiswa;
SELECT * FROM mata_kuliah;
SELECT * FROM nilai_akademik;
nilai_akademik 1
id_nilai nim kode_mk nilai_tugas nilai_uts
1 2024001 IF101 85 80
2 2024002 IF101 75 70
2 row(s) fetched - 0.001s, on 2026-11-03 at 16:35:5
Delete current row
AI Chat
Get insights based on your data: ask AI to build a query for your database or upload a file you want to analyze.
4:36 PM
11/3/2026



7. Membuat User

```
CREATE USER 'user_dosen'@'localhost' IDENTIFIED BY 'Dosen123';
CREATE USER 'user_mahasiswa'@'localhost' IDENTIFIED BY 'Mahasiswa123';
```

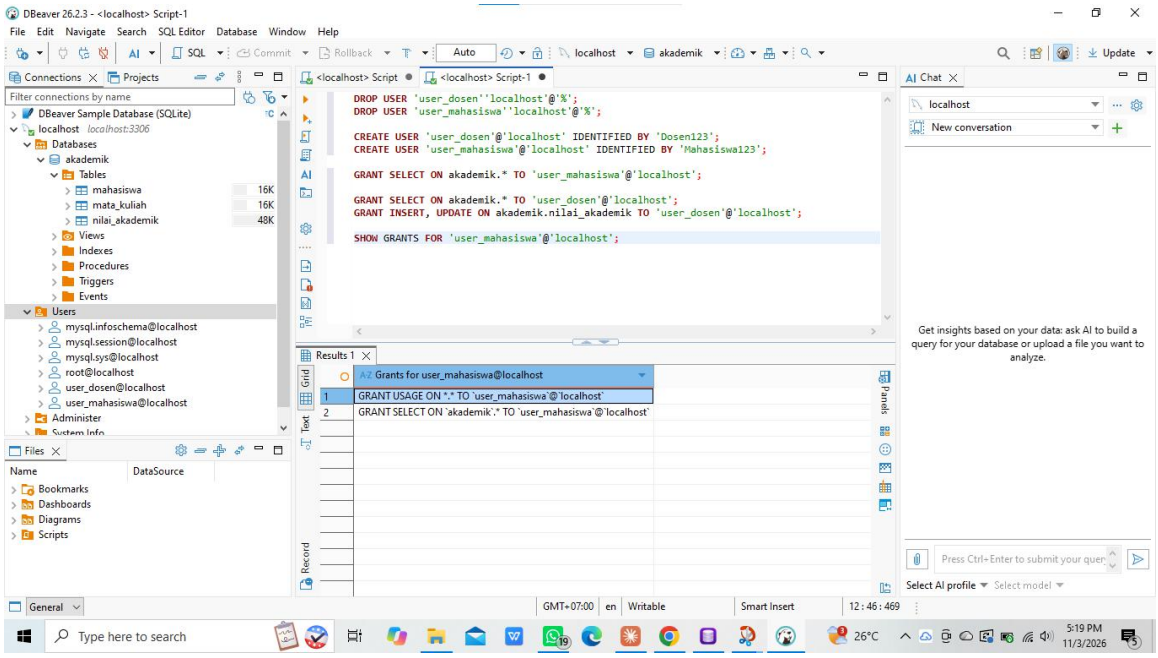
Hasil: dua user berhasil dibuat.

8. Hak Akses Mahasiswa

```
GRANT SELECT ON akademik.* TO 'user_mahasiswa'@'localhost';
SHOW GRANTS FOR 'user_mahasiswa'@'localhost';
```

Hasil SHOW GRANTS:

```
GRANT USAGE ON *.* TO `user_mahasiswa`@`localhost`
GRANT SELECT ON `akademik`.* TO `user_mahasiswa`@`localhost`
```

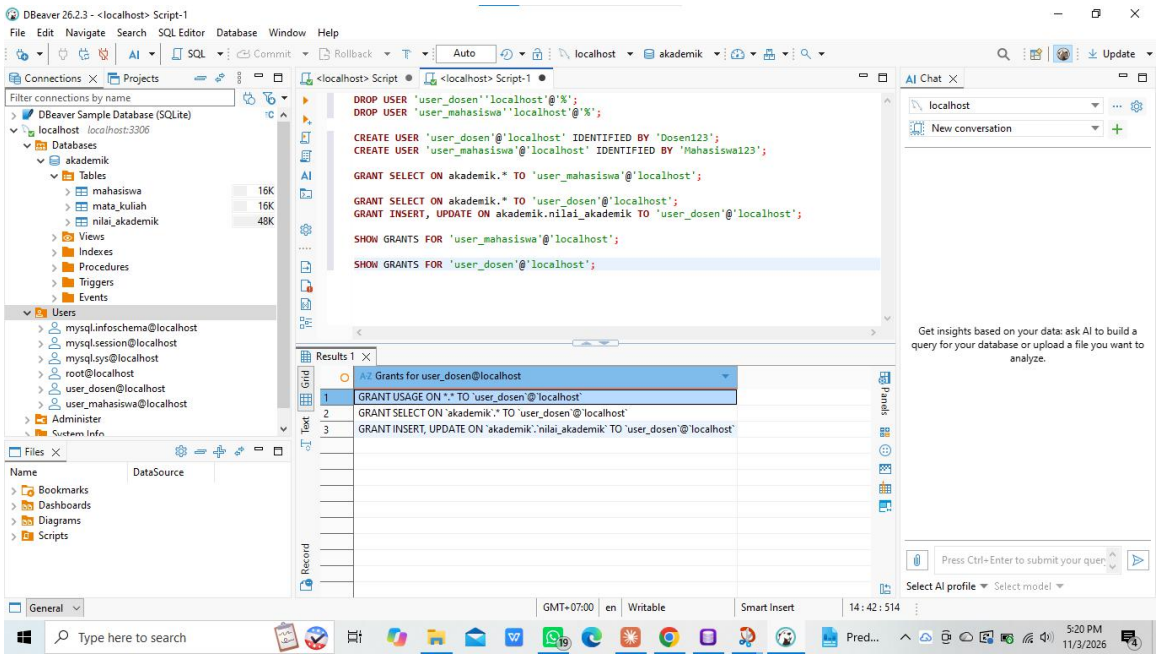


9. Hak Akses Dosen

```
GRANT SELECT ON akademik.* TO 'user_dosen'@'localhost';
GRANT INSERT, UPDATE ON akademik.nilai_akademik TO 'user_dosen'@'localhost';
SHOW GRANTS FOR 'user_dosen'@'localhost';
```

Hasil SHOW GRANTS:

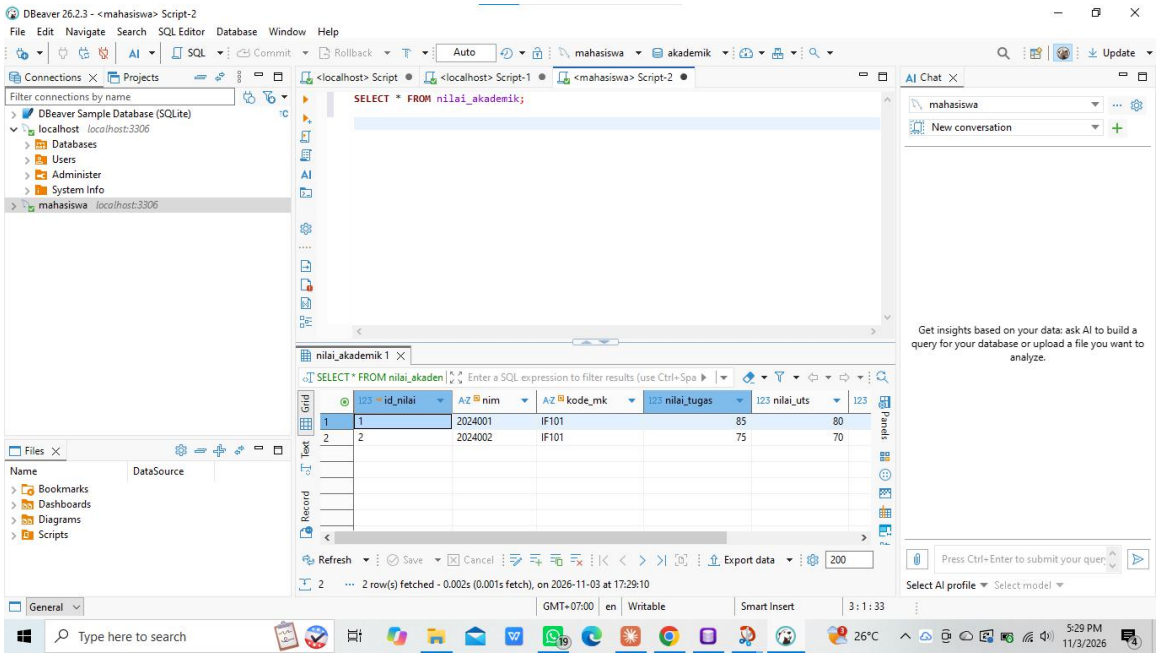
```
GRANT USAGE ON *.* TO `user_dosen`@`localhost`
GRANT SELECT ON `akademik`.* TO `user_dosen`@`localhost`
GRANT INSERT, UPDATE ON `akademik`.`nilai_akademik` TO `user_dosen`@`localhost`
```

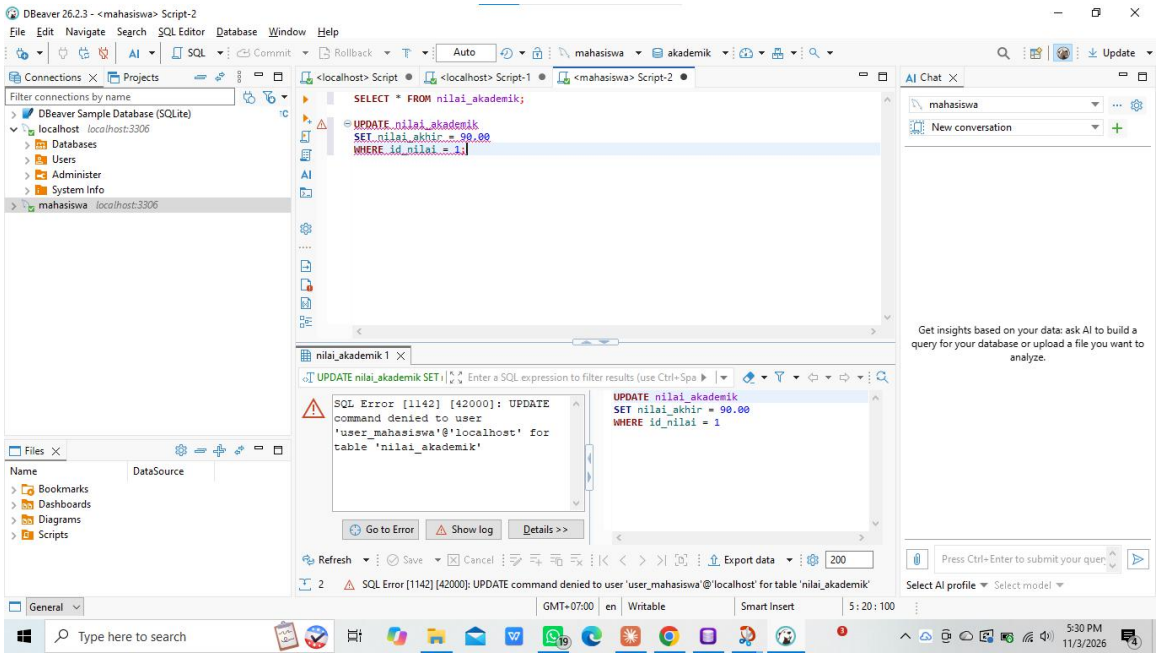


10. Pengujian User Mahasiswa

Koneksi baru di DBeaver: username `user_mahasiswa`, password `Mahasiswa123`, database `akademik`.

Perintah	Hasil
`SELECT * FROM nilai_akademik;`	Berhasil, data tampil
`UPDATE nilai_akademik SET nilai_akhir = 90.00 WHERE id_nilai = 1;`	Ditolak: *UPDATE command denied to user 'user_mahasiswa'@'localhost' for table 'nilai_akademik'*



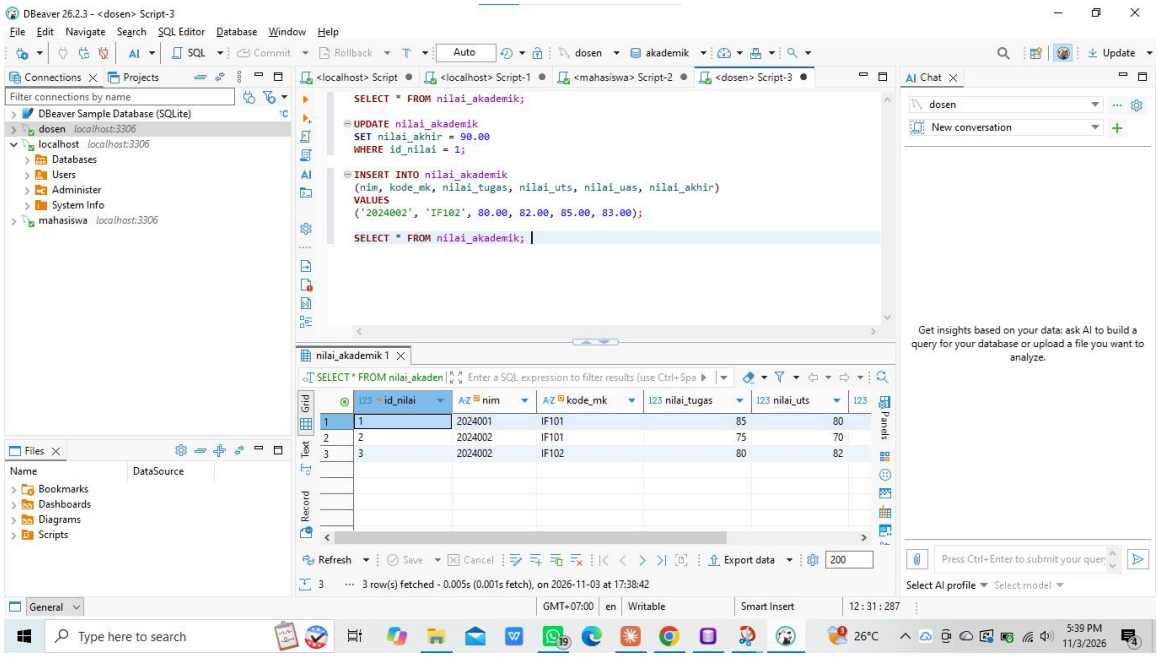


Kesimpulan: mahasiswa hanya dapat melihat data dan tidak dapat mengubah nilai.

11. Pengujian User Dosen

Koneksi baru di DBeaver: username `user_dosen`, password `Dosen123`, database `akademik`.

Perintah	Hasil
`SELECT * FROM nilai_akademik;`	Berhasil, data tampil
`UPDATE nilai_akademik SET nilai_akhir = 90.00 WHERE id_nilai = 1;`	Berhasil, nilai_akhir id_nilai 1 menjadi 90.00
`INSERT INTO nilai_akademik (nim, kode_mk, nilai_tugas, nilai_uts, nilai_uas, nilai_akhir) VALUES ('2024002', 'IF102', 80.00, 82.00, 85.00, 83.00);`	Berhasil, 1 baris baru ditambahkan



12. Perbandingan Hak Dosen dan Mahasiswa

Aktivitas	Dosen	Mahasiswa
SELECT data	Boleh	Boleh
INSERT nilai	Boleh	Tidak
UPDATE nilai	Boleh	Tidak
DELETE nilai	Tidak diberikan	Tidak diberikan

13. Akses Tabel Tertentu

```
GRANT SELECT ON akademik.mahasiswa TO 'user_dosen'@'localhost';
```

`akademik.*` berarti seluruh tabel dalam database, sedangkan `akademik.mahasiswa` hanya tabel mahasiswa.

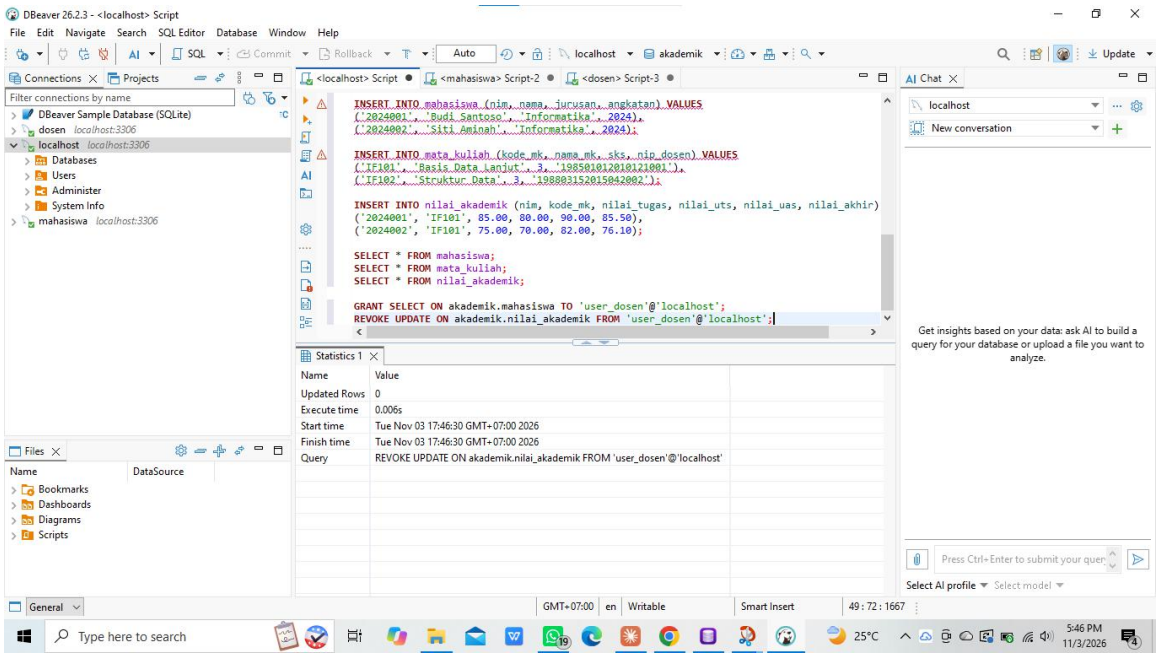
14. Mencabut Hak Akses (REVOKE)

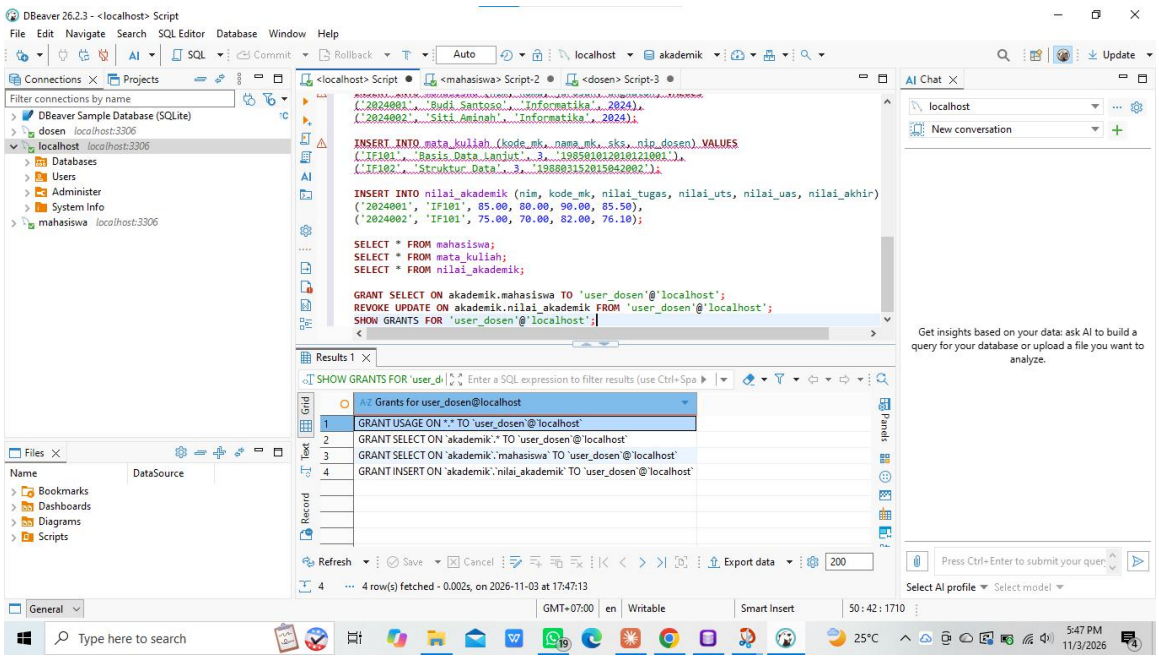
```
REVOKE UPDATE ON akademik.nilai_akademik FROM 'user_dosen'@'localhost';
SHOW GRANTS FOR 'user_dosen'@'localhost';
```

Hasil SHOW GRANTS setelah REVOKE:

```
GRANT USAGE ON *.* TO `user_dosen`@`localhost`
GRANT SELECT ON `akademik`.* TO `user_dosen`@`localhost`
GRANT SELECT ON `akademik`.`mahasiswa` TO `user_dosen`@`localhost`
GRANT INSERT ON `akademik`.`nilai_akademik` TO
`user_dosen`@`localhost`
```

Hak UPDATE pada `nilai_akademik` sudah tidak ada, sehingga dosen tidak lagi dapat mengubah nilai.





15. Kesimpulan

MySQL dapat membedakan hak akses berdasarkan pengguna. Mahasiswa hanya diberi hak SELECT sehingga hanya dapat melihat data. Dosen diberi SELECT pada database serta INSERT dan UPDATE pada tabel `nilai_akademik` sehingga dapat mengelola nilai. Perintah REVOKE berhasil mencabut hak UPDATE dosen. Prinsip pentingnya: hak akses diberikan sesuai peran dan kebutuhan pengguna.